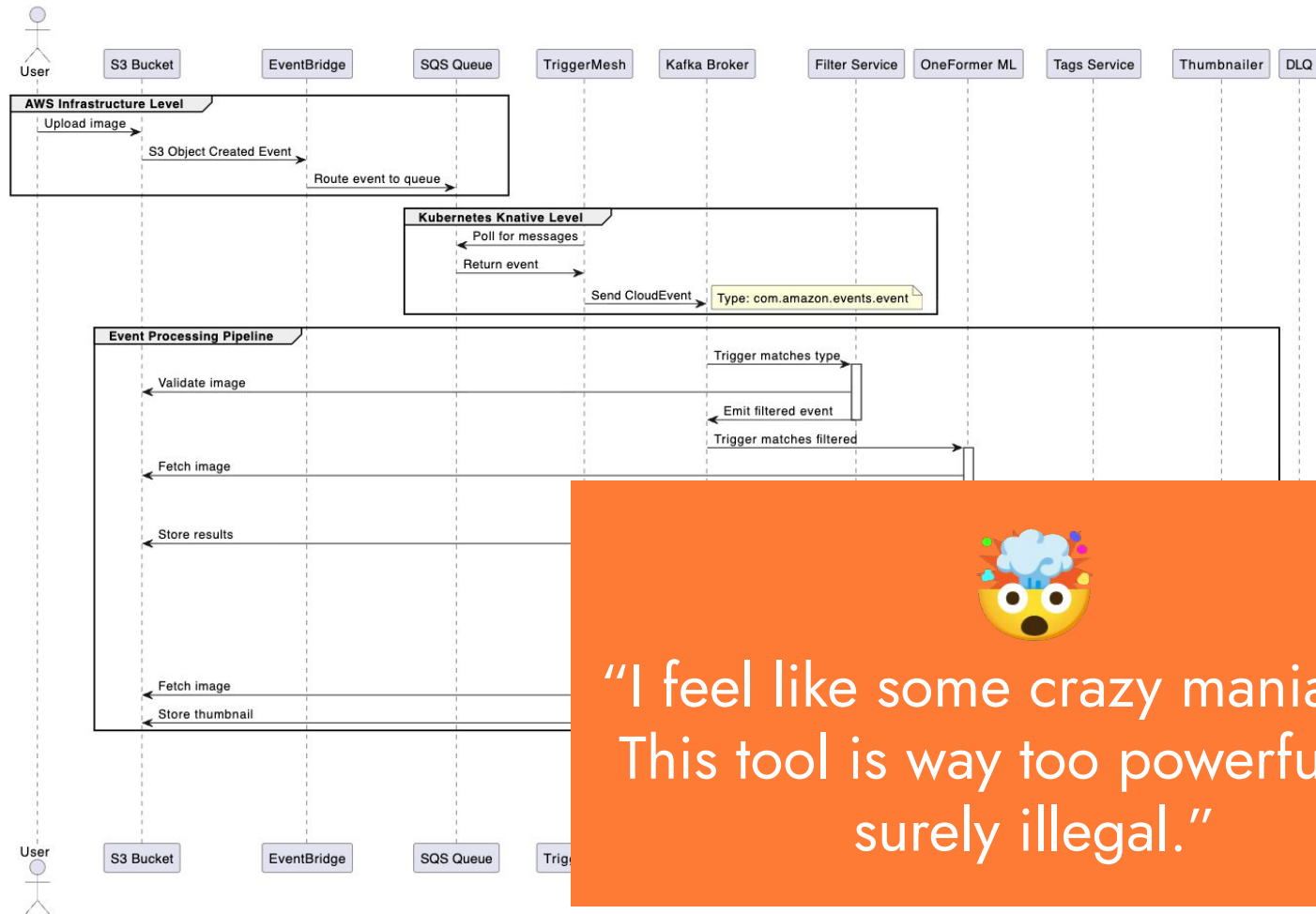


Agentic Coding and Software Quality

Experience, Early Evidence, and Open Questions



Dr. Andreas Wilhelm



"I feel like some crazy maniac [...]
This tool is way too powerful and
surely illegal."



This is the r
with cl

Claude v

The CPrep

Browser window showing the Jira issue page for "Rust scanner/parser".

URL: cqse.atlassian.net/browse/TS-37831?search_id=a8887ef5-26ef-4c72-90d1-7bbade5c0f93&referrer=quick-find

Spaces / Teamscale / TS-32203 / TS-37831

Rust scanner/parser

+ -

Key details

Main HubSpot

Description

User Story

In order to use Teamscale for a Rust codebase, as a software engineer, I can analyze and parse Rust code using a dedicated scanner and parser.

Context

As Rust continues to gain popularity, particularly for system-level and concurrent computing programming, the necessity to incorporate Rust into Teamscale becomes imperative.

We already attempted to implement a Rust Scanner/Parser around 2016 which proved to be

Details

Original estimate: 2w 2d

Time tracking: 4w 1d 6h 12m logged

Fix versions: Teamscale 2026.5

Teamscale

Project name	Findings balance	Test Gap
Teamscale	103 added, 4 in changed code, 2 removed	10.44%



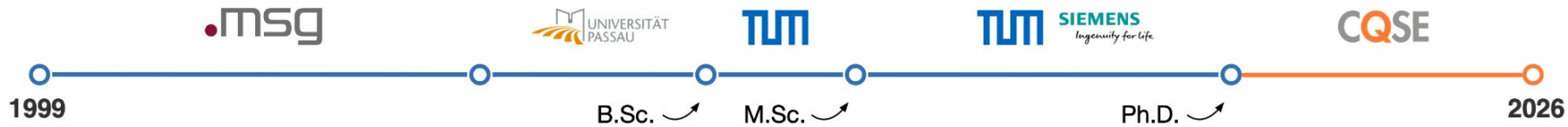
Dies ist eine automatisierte und vorläufige Beurteilung (KI-Ersteinschätzung). Nur der Befund des Radiologen ist endgültig!





Disclaimer

AI-models evolve almost every week
Today's best practices expire very fast
We are constantly learning





 **Teamscale**
Software quality platform



Consultancy and support
for quality improvement

CQSE



25+ PhDs in Software Engineering



Own research



Close contact to academia



VER**SICHER**UNGS
KAMMER
BAYERN

LV 1871

BayBG

...



SIEMENS

NORDEX

ASMPT

SEIDENADER
KÖRPER SOLUTIONS

ROSEN

...



Vibracoustic

FERNRIE



SEW
EURODRIVE

...



TTTech

SALZGITTER
DIGITAL SOLUTIONS
Ein Unternehmen der Salzgitter Gruppe

tebis

CAD/CAM MES
Software & Services

dm
TECH

...

...

SW/M



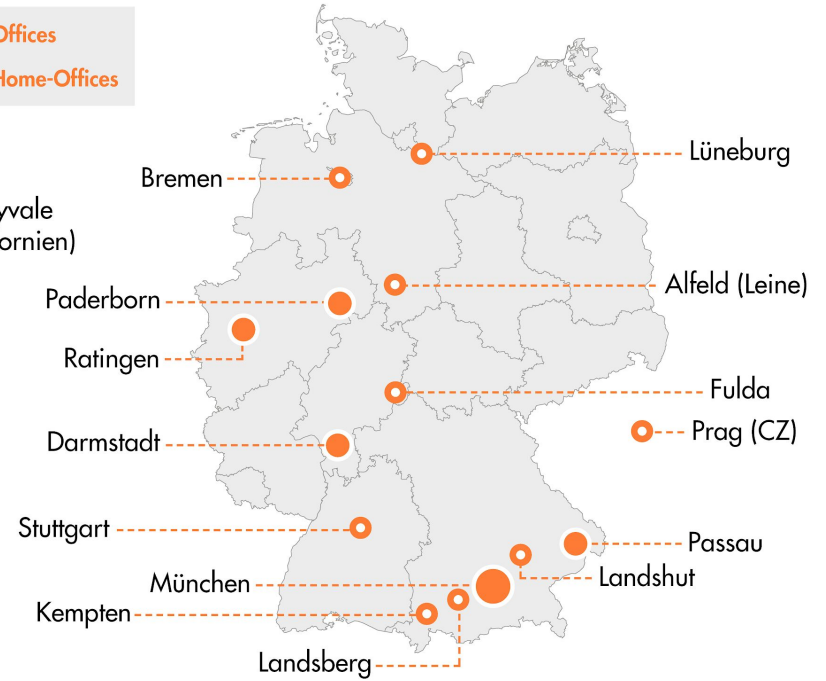
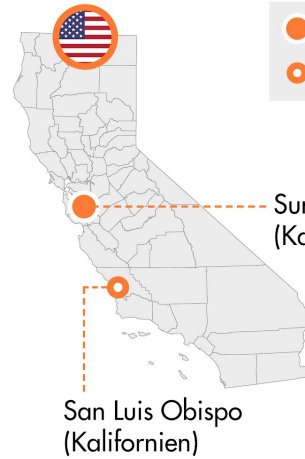
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CQSE

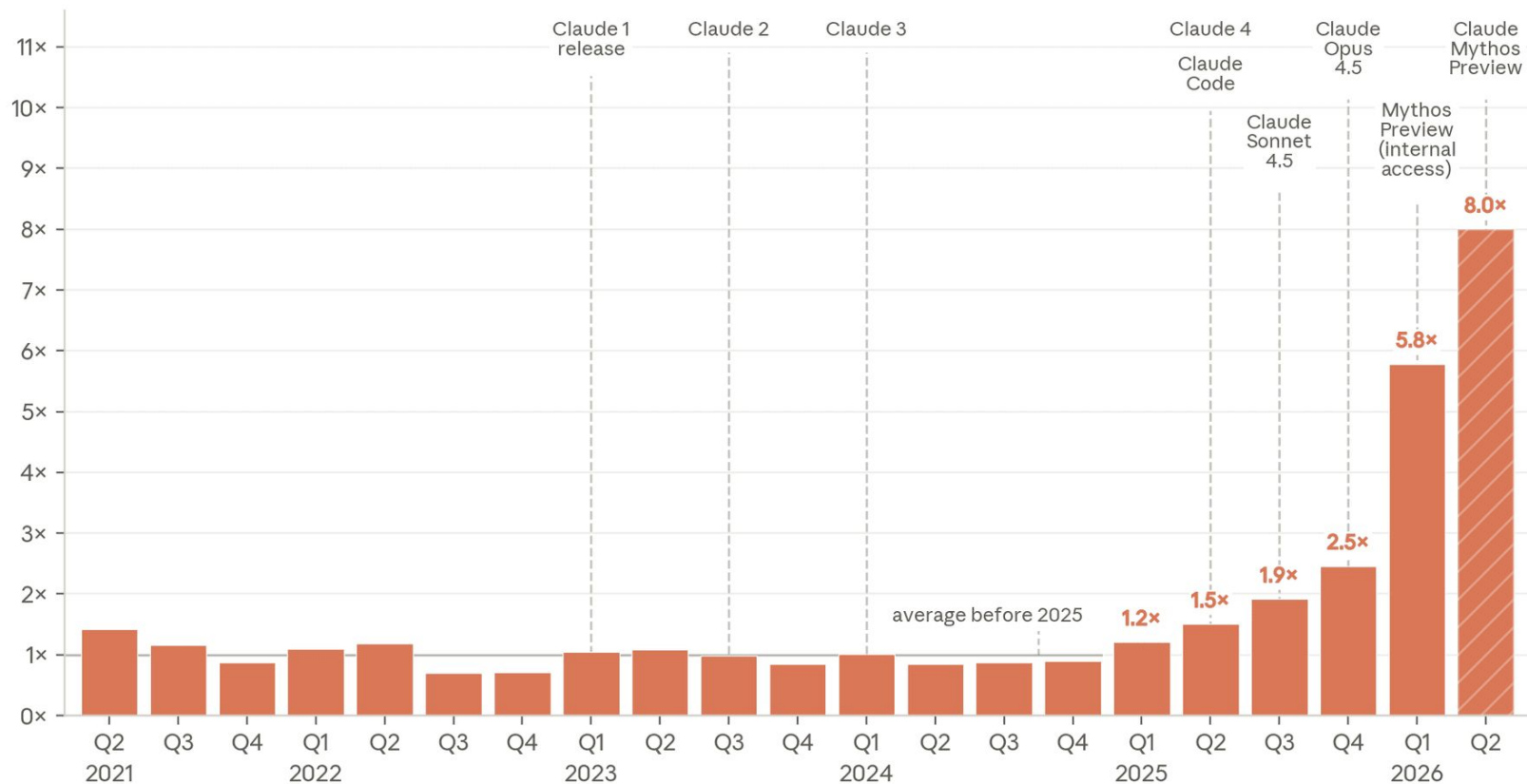


Agentic Coding

Levels of AI Code Agent Autonomy

Level	Name	Behavior	Typical UX
0	Static Tooling	Rule-based helpers like linters or formatters—no AI	Tooltips, sidebars
1	Token-Level Completion	Predicts next token or word based on local context	Inline
2	Block-Level Completion	Completes entire lines or functions	Inline
3	Intent-Based Chat Agent	Chat-driven assistant suggests changes	Chat UI + Diff
4	Local Autonomous Agent	Receives a feature description, edits files, runs tests—ITERATES in your workspace	Chat UI, Diff, Terminal, Planner UI
5	Fully Autonomous Dev Agent	Plans and completes tasks end-to-end	Agent Dashboard

Code contributed per person, by quarter



Each bar is the average, over the days in that quarter, of lines of code merged per active contributor — shown as a multiple of the pre-2025 average. The hatched final bar is a partial quarter: it averages only the days observed so far, not a full quarter. Dashed lines mark public announcement dates. Per-PR line counts are capped at the 99th percentile; “active contributor” means a distinct author in the trailing twelve months.



AI

Jensen Huang says he would be 'deeply alarmed' if his \$500,000 engineer did not consume at least \$250,000 of tokens

By [Lee Chong Ming](#)

+ Follow

```
while true
do
  claude -p "Task $RANDOM: " \
    "Write a 1000 word essay on AI."
done
```



THE ACCELERATION WHIPLASH

Engineering throughput is up. However, bugs, incidents, and rework are rising faster.

Analysis of two years of data from 22,000 developers on the real-world impact of AI adoption.

OUTPUT IS UP. NOT ALL OF IT STICKS.

Impacts of high AI adoption on engineering throughput



Epic

A sizable feature or capability, hundreds of tasks



+66.2%

Epics completed per developer



Task

Individual work item, may span multiple PRs



+33.7%

Task throughput per developer

Within this, tasks with an associated PR grew **+210%** per team. Code-specific work accelerated far faster than overall task volume.

CODE IS GETTING LARGER AND MORE COMPLEX

Impacts of high AI adoption on code complexity

AI ADOPTION

LOW



HIGH

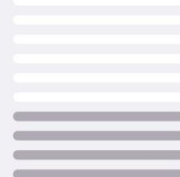


+59.7%

files edited per PR

AI ADOPTION

LOW



HIGH



+51.3%

average PR size

DEVELOPER
FOOTPRINT/MONTH



+149.9%

files touched per developer



+11.7%

repositories touched per developer

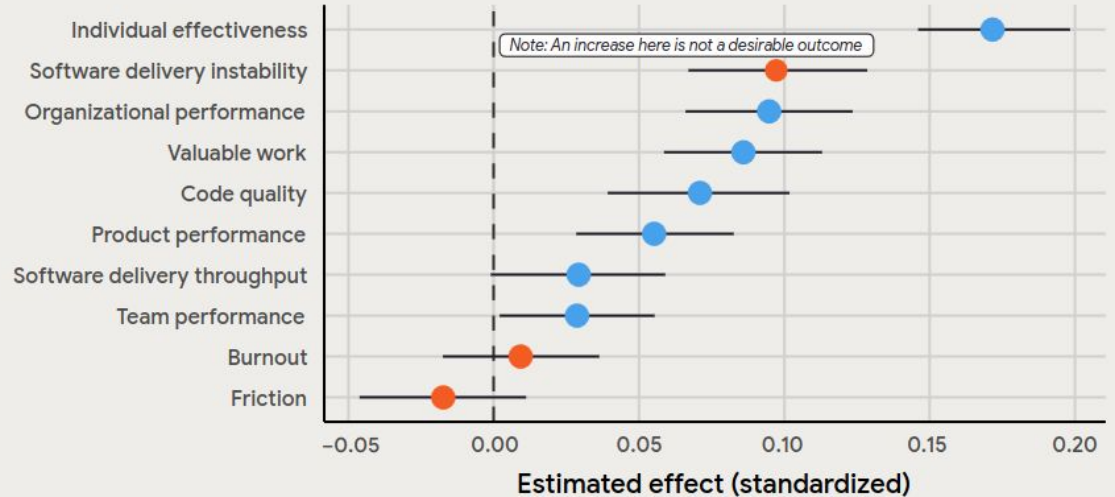
The ROI of AI-assisted Software Development

Google Cloud



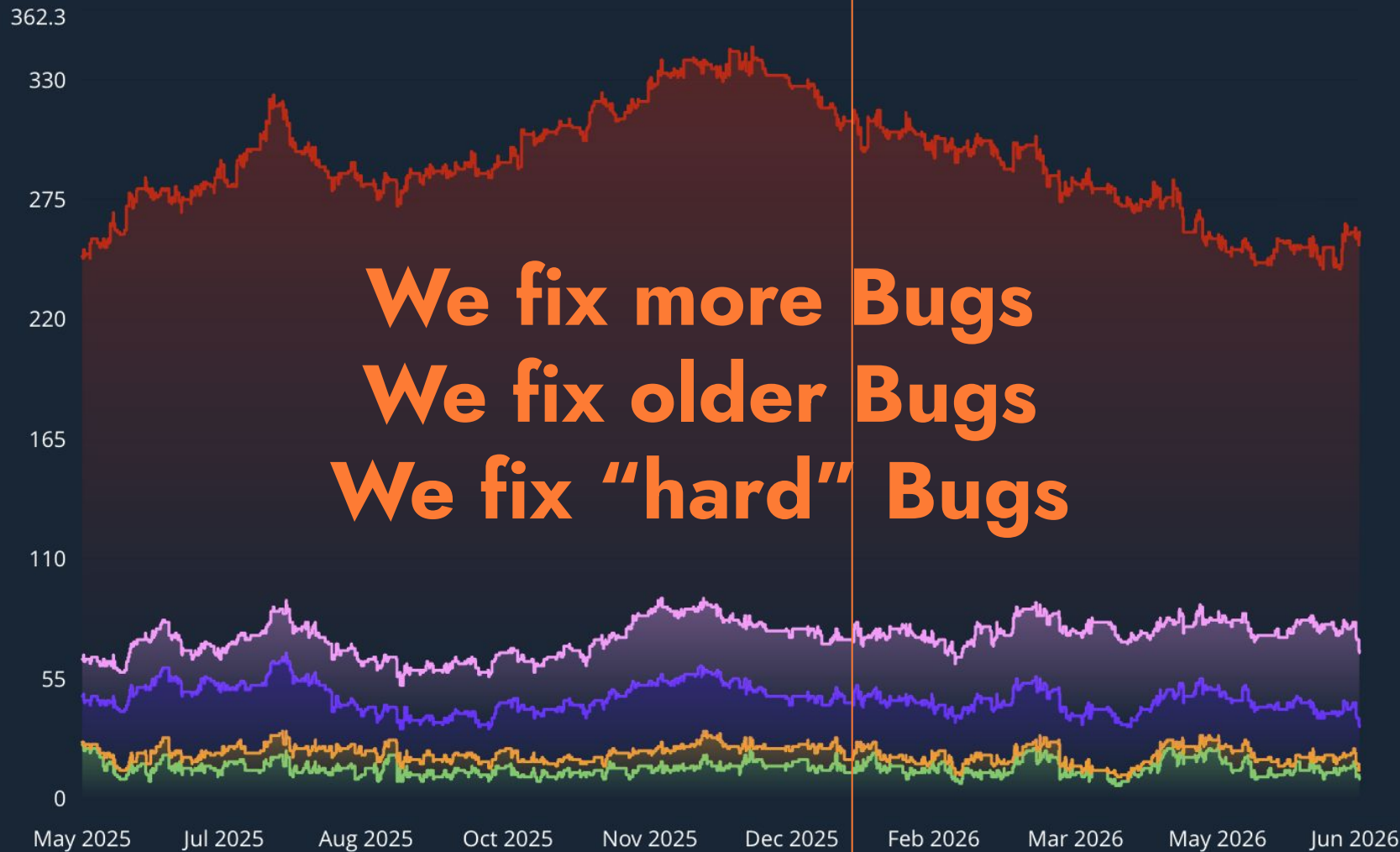
The landscape of AI's impact

Estimated effect of AI adoption on key outcomes, with 89% credible intervals

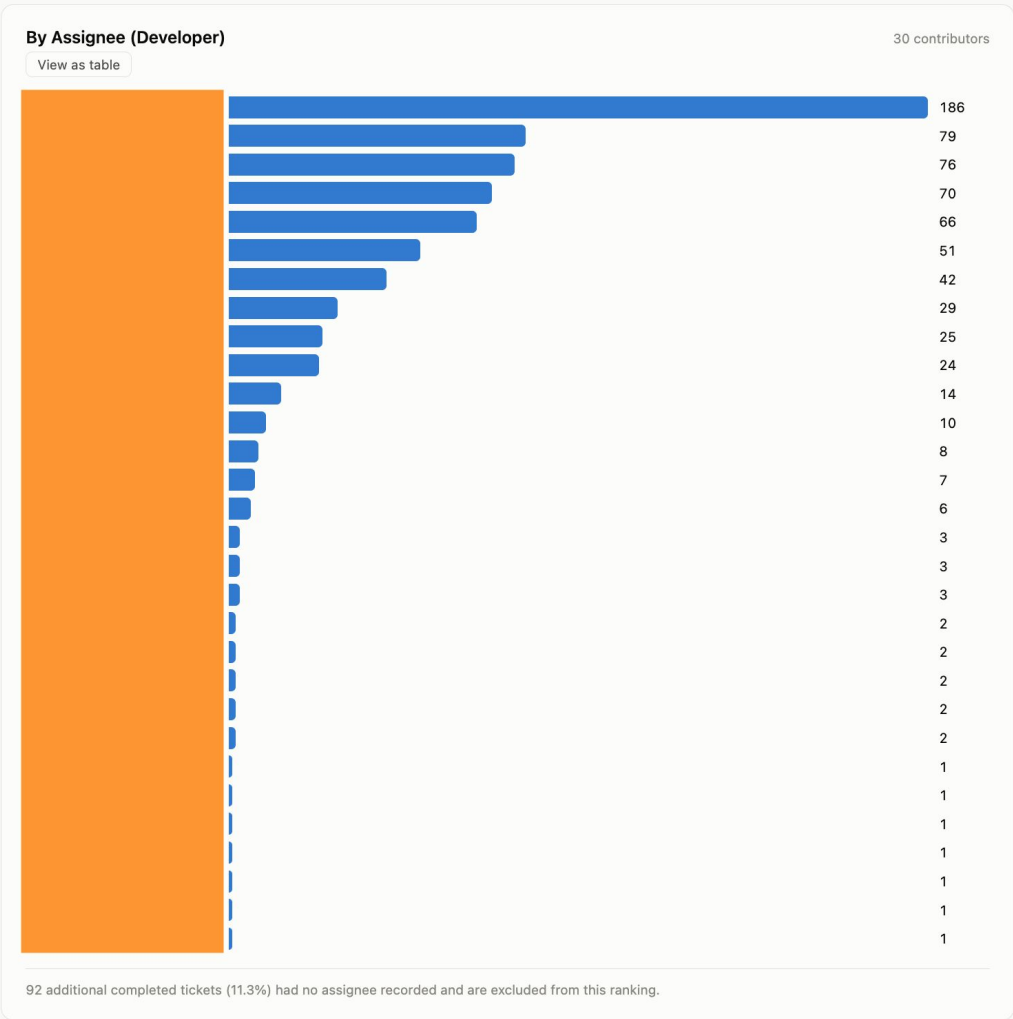


For outcomes in orange (such as burnout), a negative effect is desirable.

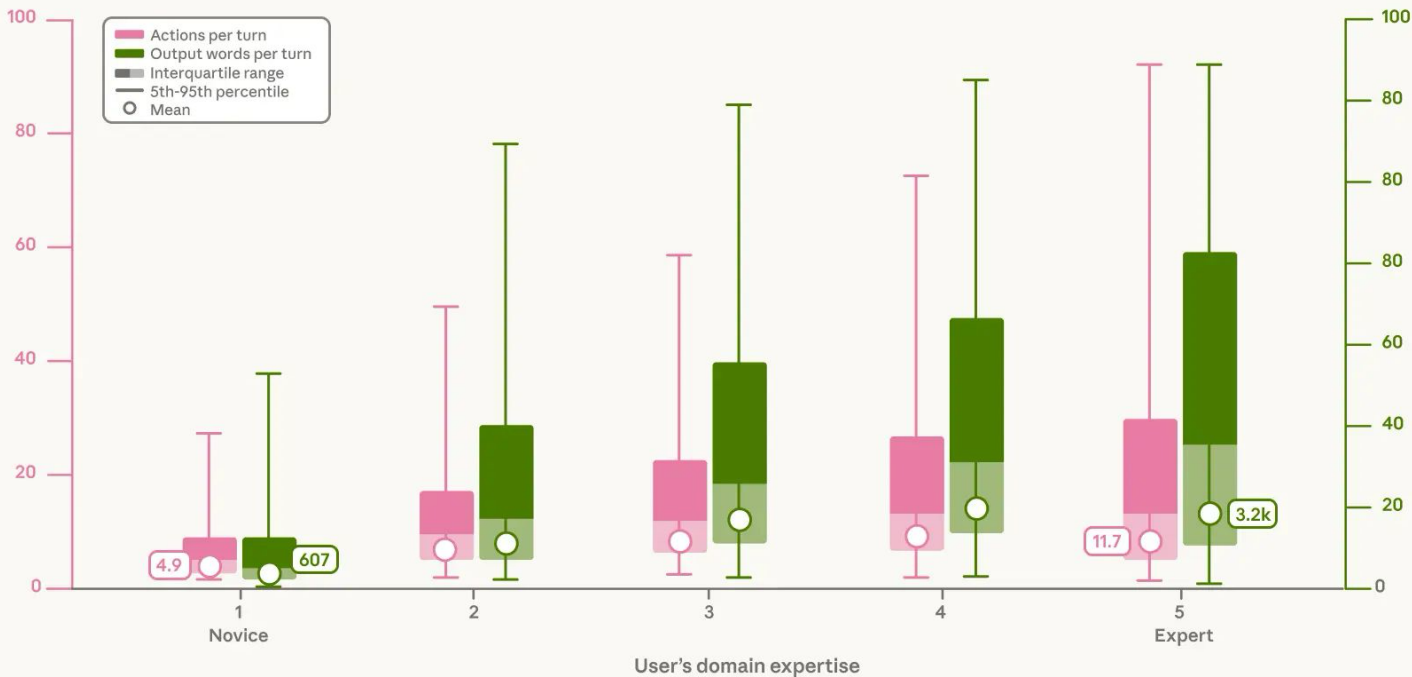
Figure 4: Standardized estimated effects of AI adoption



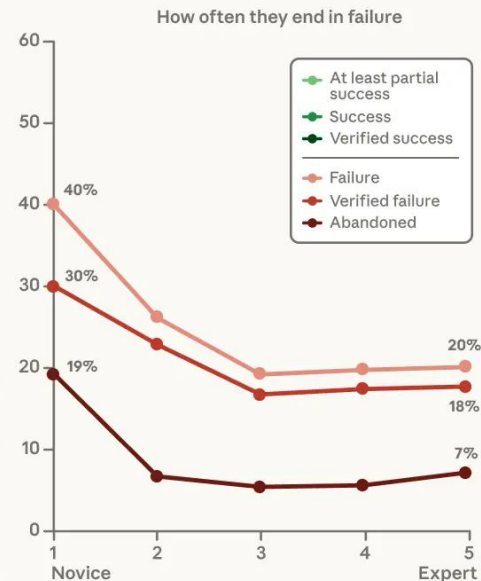
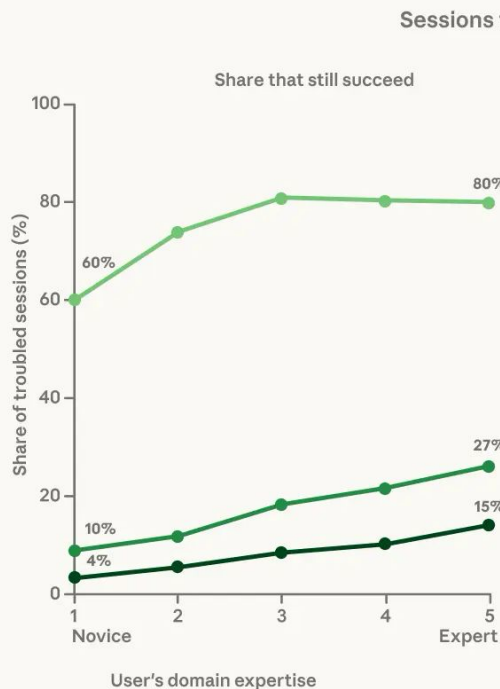
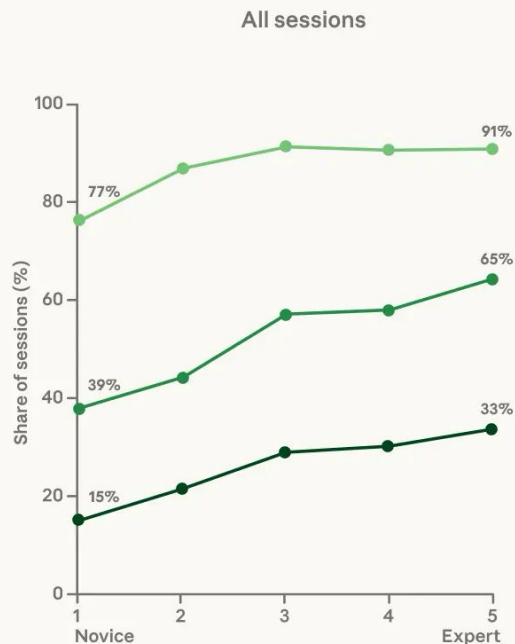
“Strong engineering foundations
amplify AI's benefits and offer
protection against its downsides.”



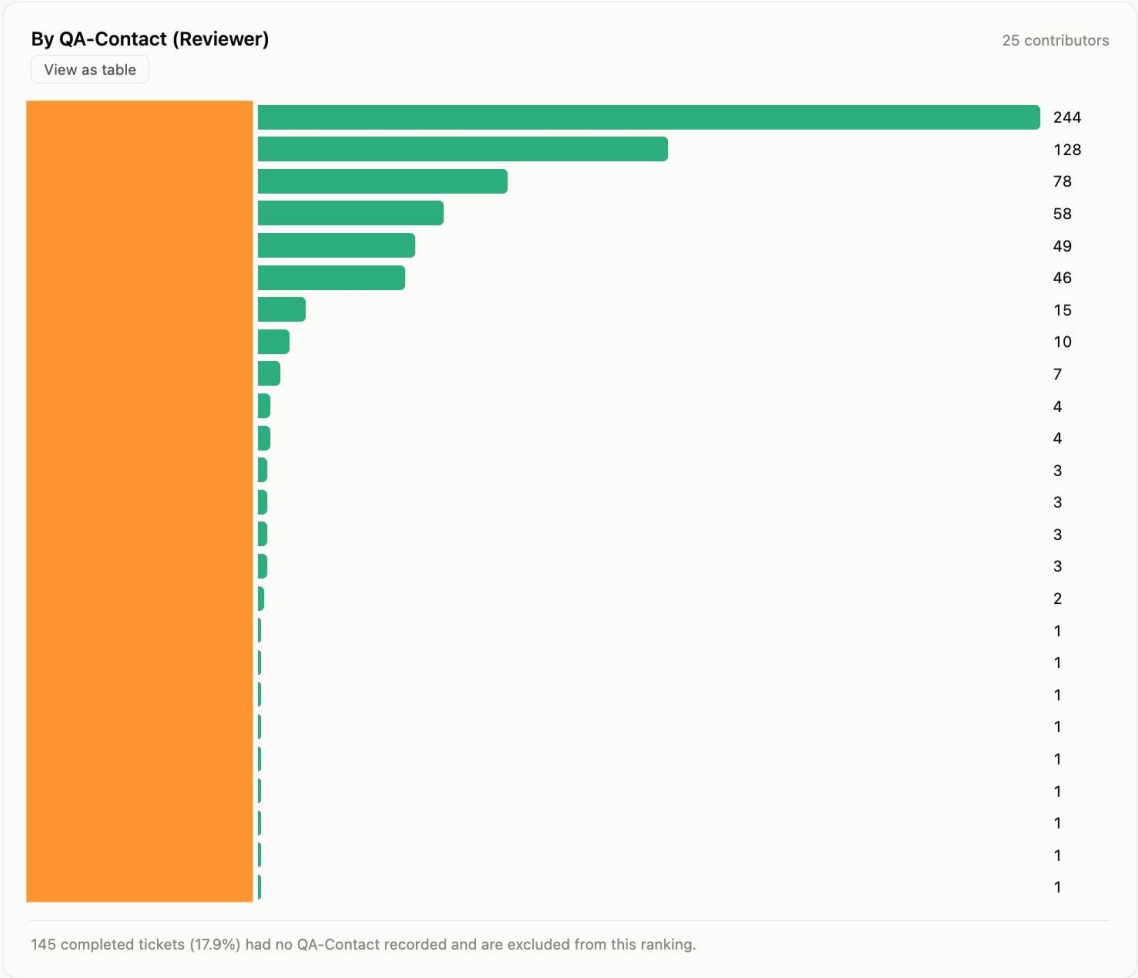
Claude does more per prompt for more expert users



Expertise and how sessions end

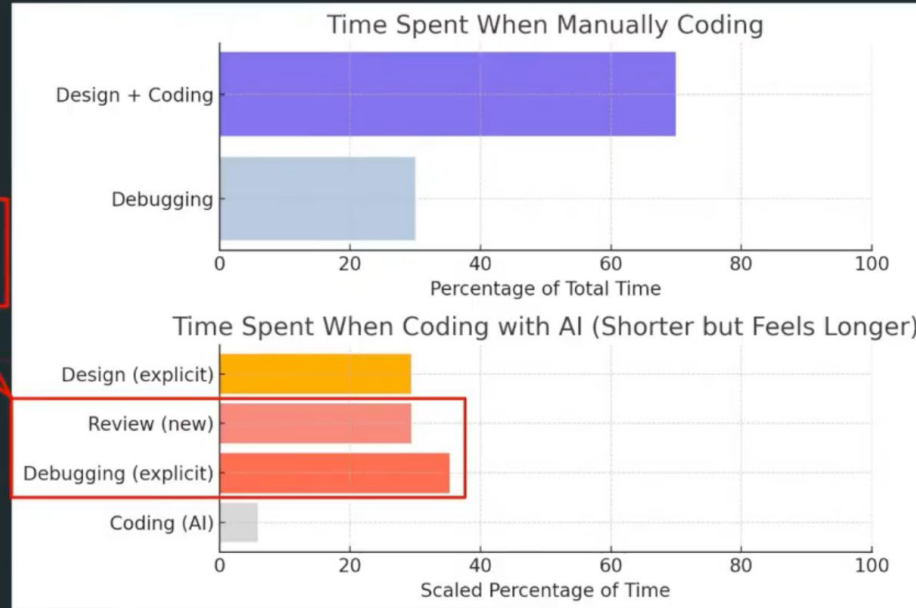


“The most experienced people in your organization are being buried. We call it the senior engineer tax.”



How we spend our time while coding

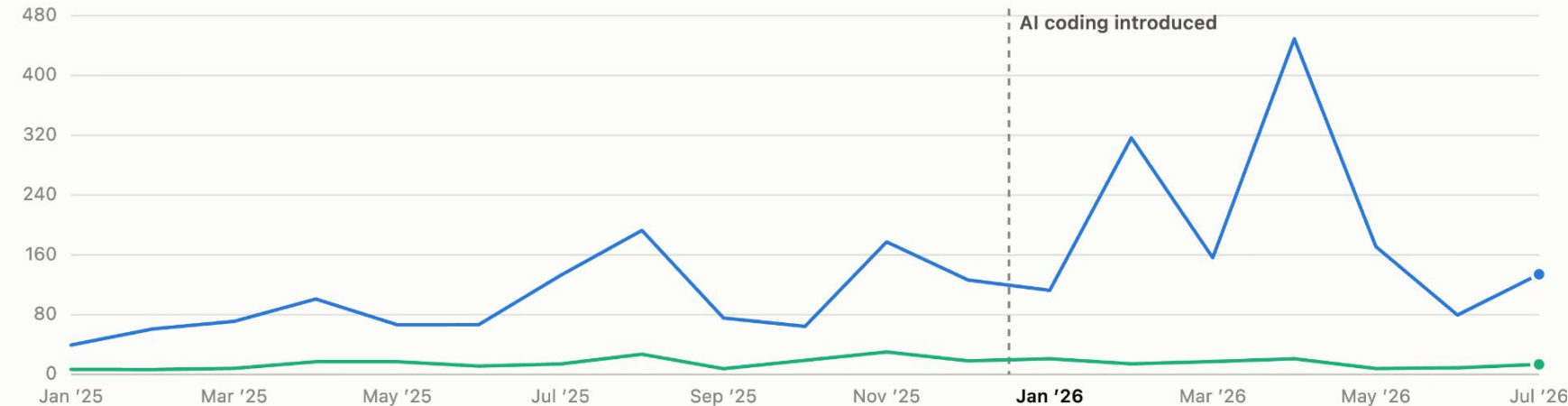
Here is where we can save time!



Age at resolution, by month

Average vs. median days between ticket creation and resolution

Average age (days) Median age (days)



Thinking—Fast, Slow, and Artificial:

How AI is Reshaping Human Reasoning

and the Rise of Cognitive Surrender

Steven D. Shaw
The Wharton School of the University of Pennsylvania
3733 Spruce Street, Philadelphia, PA 19104
Email: shawsd@wharton.upenn.edu

Gideon Nave*
The Wharton School of the University of Pennsylvania
3733 Spruce Street, Philadelphia, PA 19104
Email: gnave@wharton.upenn.edu



Effective Learning requires “Deep Practice”.

“You must stretch, fail, notice the mistake,
correct it, and repeat.”

‘A helluva good read’
Tom Peters, author of *In Search of Excellence*

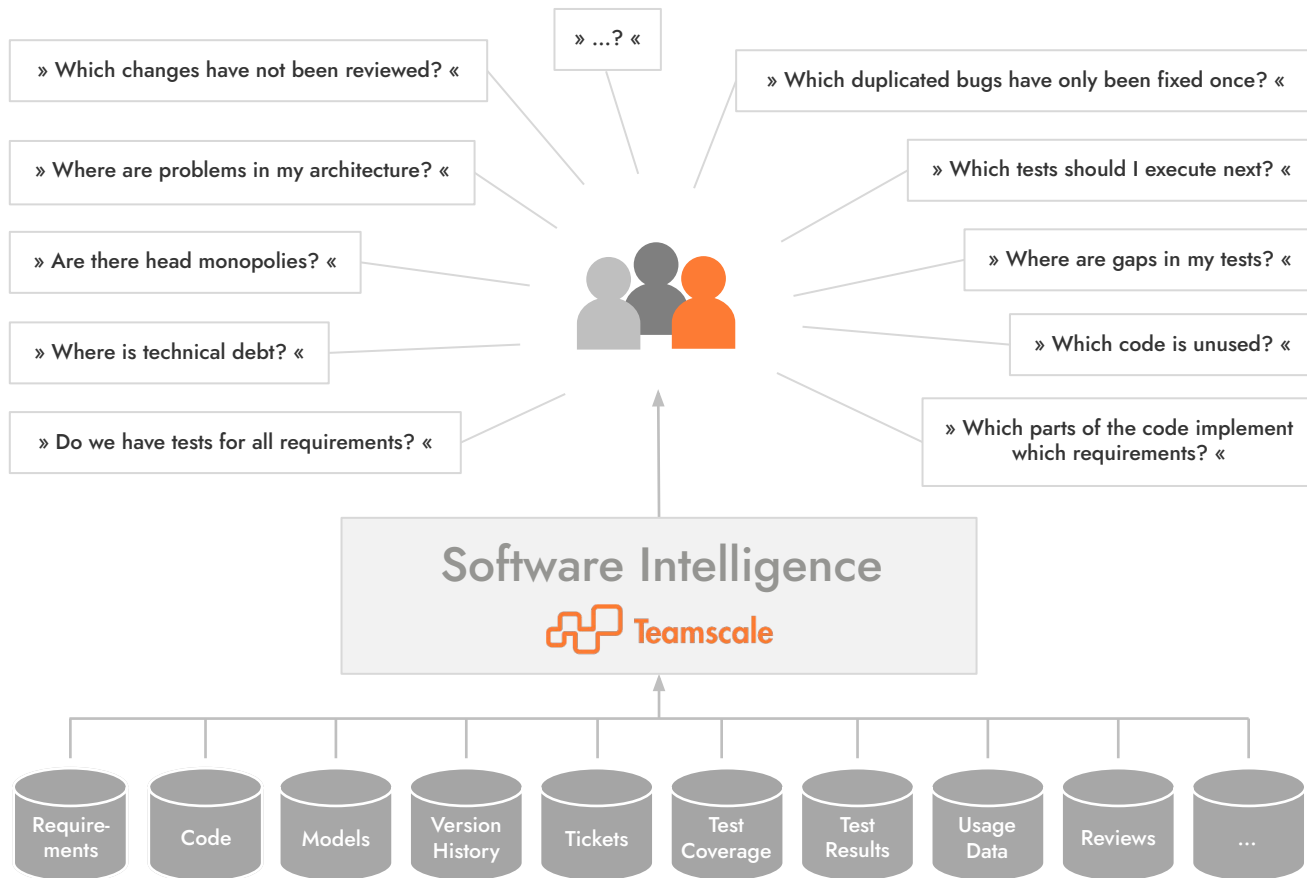
DANIEL COYLE
**THE
TALENT
CODE**

GREATNESS ISN'T BORN.
IT'S GROWN.

From the *New York Times* bestselling author of
The Culture Code

Teamscale

The Software Quality Platform For Great Teams

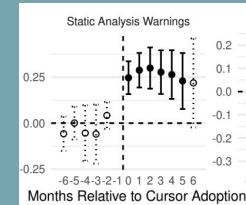


What are facts about AI generated code?

“The quality of AI generated code is so good, we no longer need additional quality measures.”

There is evidence that AI coding erodes quality.

He et al.: Speed at the Cost of Quality. MSR 2026



More and **more AIs** generating average code lead to **higher QA demand**:
QA becomes the bottleneck for scaling software development!

What are facts about AI generated code?

"Softw

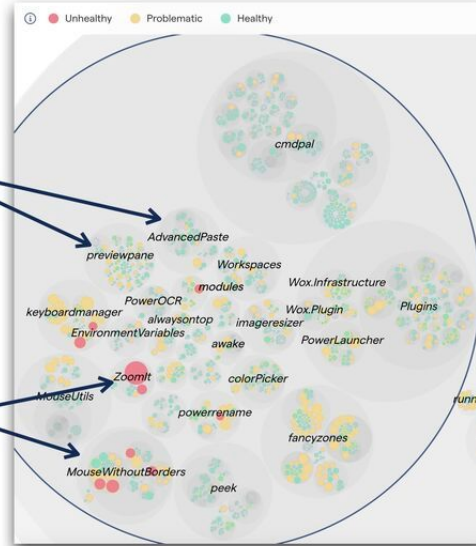
nges.

There

code.

Healthy code == AI-friendly 🙌

Unhealthy code == danger zone:
increased break rates 😬



You need Adam Tornhill on LinkedIn

Example: Code Health visualization of the PowerToys repo via CodeScene.

omation.

Better keep the code clean, so that AI and humans can still understand it.

Social Aspects



Social Aspects

Of course Claude should modularize this function.

I feel much better now that Claude closed these test gaps.



Thank you Teamscale for pointing out room for improvement.



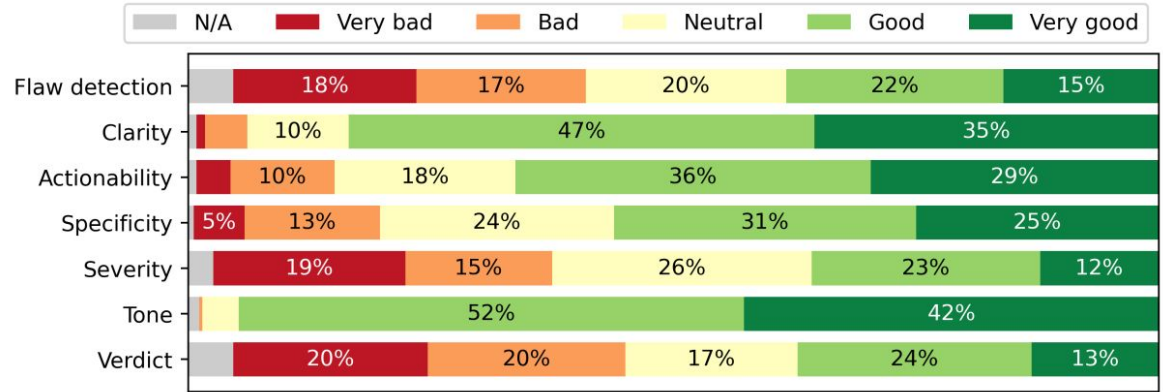
SCHOOL OF COMPUTATION, INFORMATION
AND TECHNOLOGY — INFORMATICS

TECHNISCHE UNIVERSITÄT MÜNCHEN

Bachelor's Thesis in Informatics

Using AI for Generating Code Review Comments

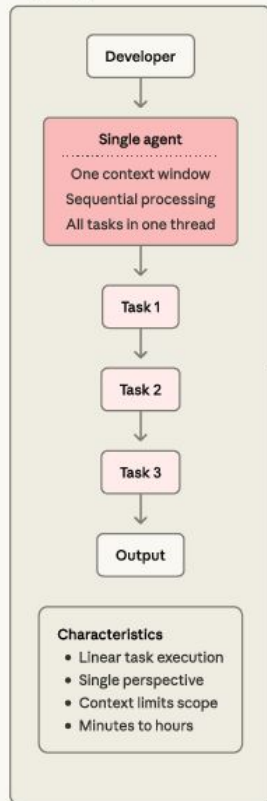
Dominik Weißenseel



“LLMs are very good in convincing non-experts
(=junior developers?!) in the existence of quality issues”

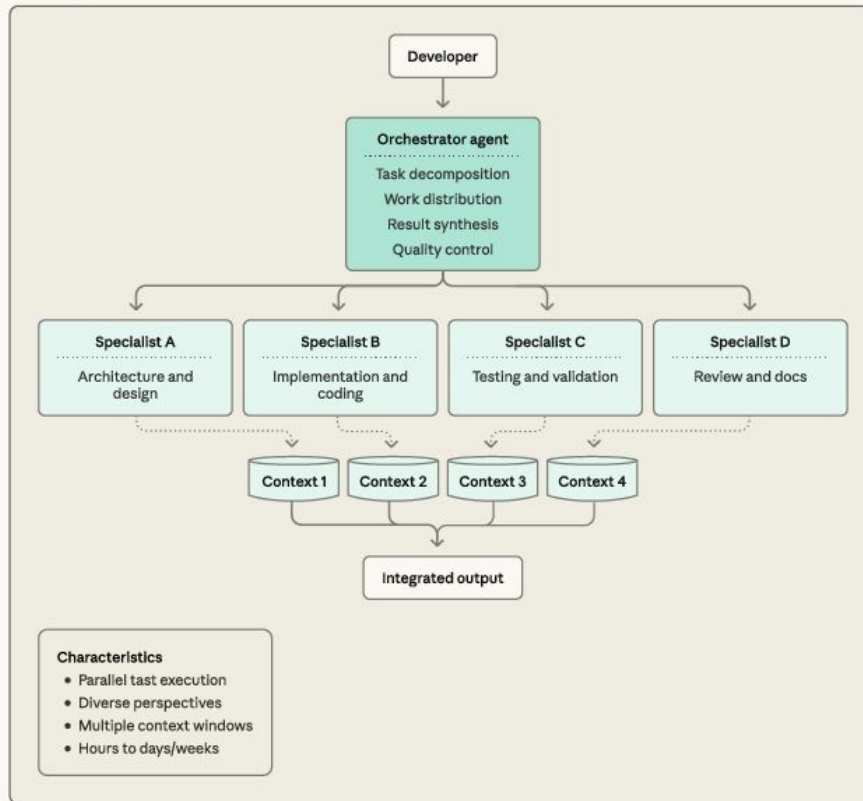
Coding agent architectures: From single agents to coordinated teams

Single agent architecture



Evolution
Task horizons
expand from
minutes to weeks

Multi-agent hierarchical architecture



Prompt Engineering

Single-call Optimization Optimize the **model input**



Instruction & Role

*Define task, persona
and constraints*



Example & Format

*Define task, persona
and constraints*



Prompt Iteration

*Refine wording from
observed failures*

Context Engineering

Multi-step Context Optimization Optimize **what the model sees**



Retrieval & Loading

*Bring in relevant files, docs
and tool outputs*



Memory & State

*Track progress, decisions,
and prior steps*



Compression & Filtering

*Keep useful context,
and remove stale information*

Harness Engineering

System-level Optimization Optimize **how the model runs**



Tools & Environment

*Run action safely and
expose callable tools*



Orchestration

*Plan, act, verify, retry,
or delegate*



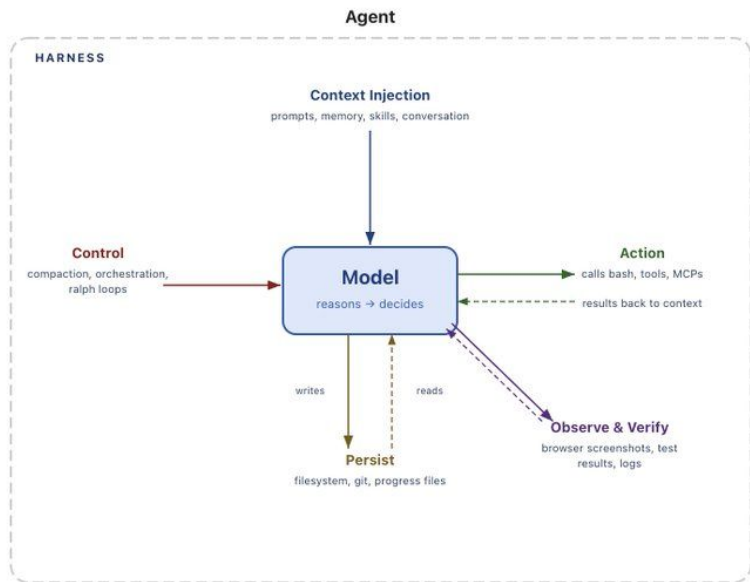
Verification & Evaluation

*Trace behavior, measure cost,
and check results*



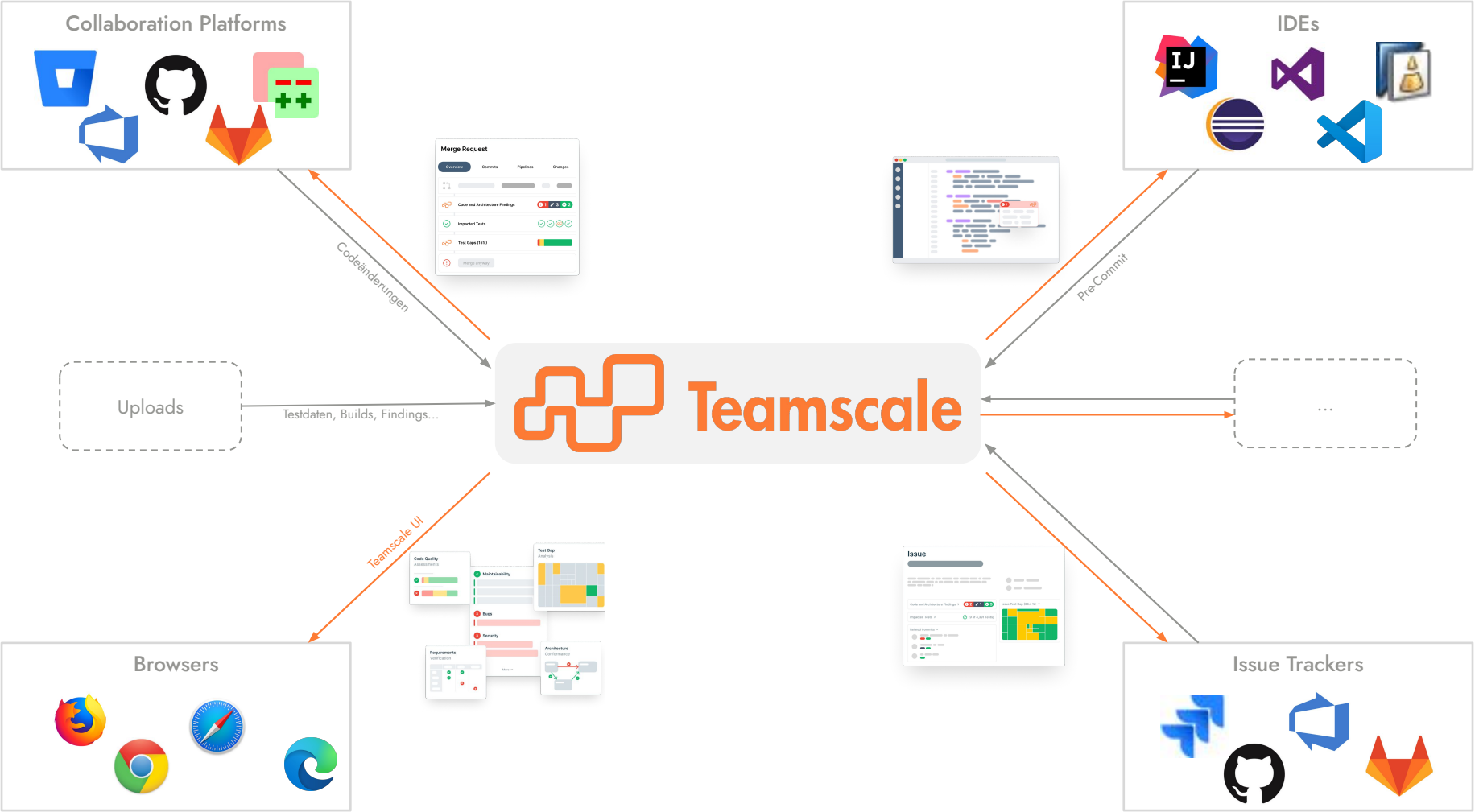
Governance & Security

*Enforce permissions, approvals
and audit rules*



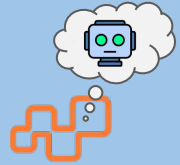
Desired Agent Behavior	What the Harness Adds
Work with real data durably	Filesystem + Git
Write & Execute Code	Bash + Code Execution
Safe Execution + Default Tooling	Sandboxed Environments + Tooling
Remember and access new knowledge	Memory Files + Web Search + MCPs
Maintain performance over long contexts	Compaction + Tool Offloading + Skills
Complete long horizon work	Ralph Loops + Planning + Verification

Each harness feature derives from a behavior the model can't deliver on its own



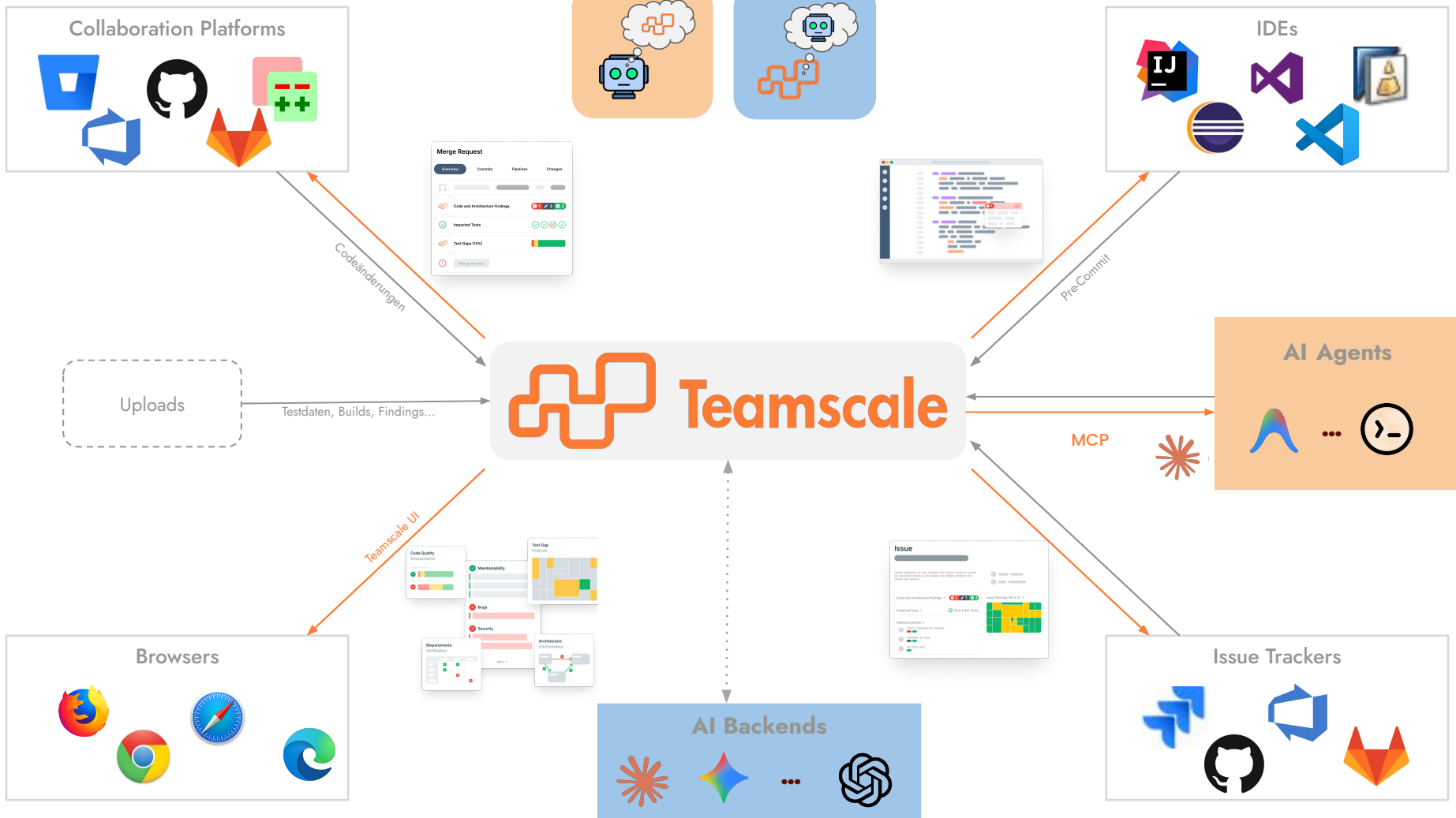
Use Cases involving AI

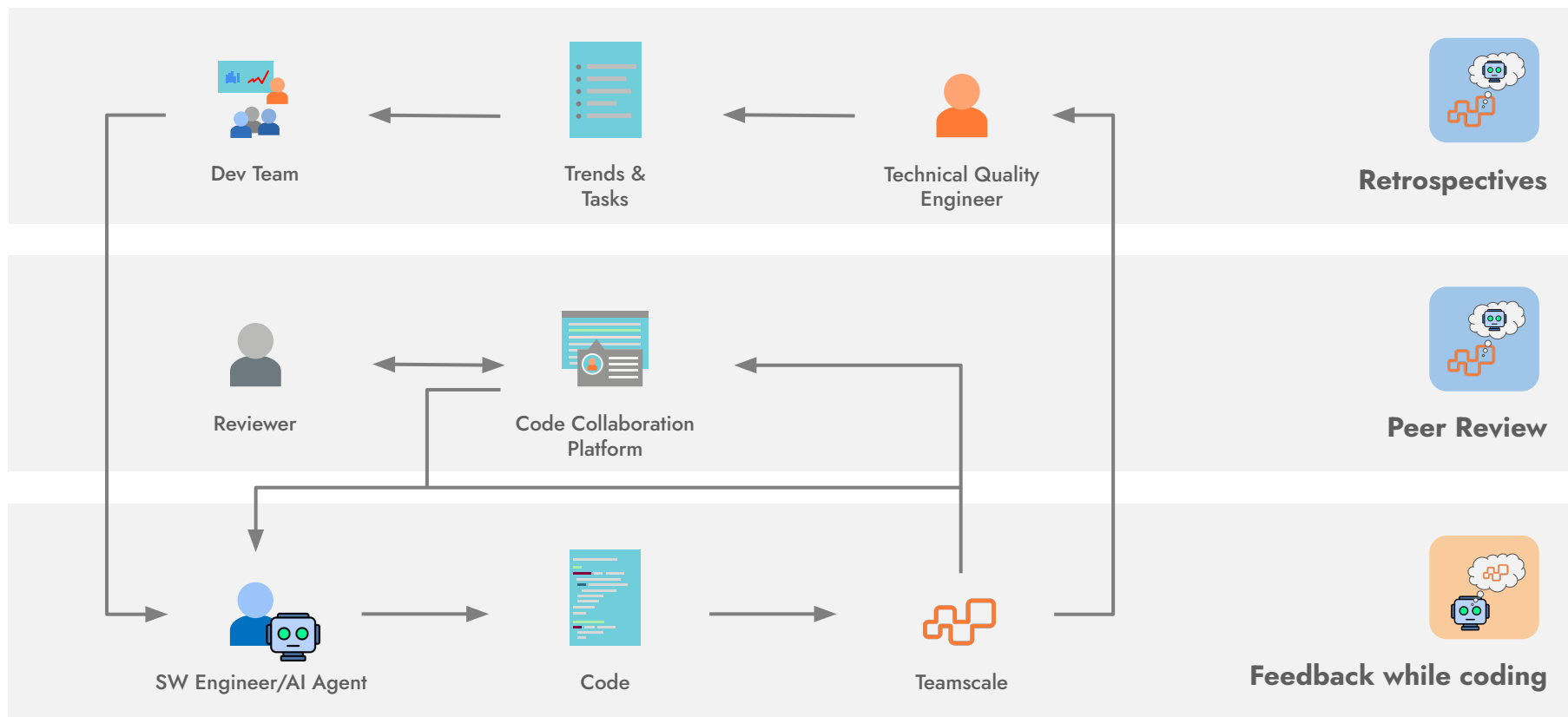
- Get refactoring suggestion for a finding from an AI
- Have an AI review code changes in a merge request
- Ask an AI to explain and summarize content a user sees in Teamscale
- ...



- Ask AI agent to clean up Teamscale findings on local changes
- Ask AI agent to execute the tests that run through local changes
- Ask AI to generate tests for remaining test gaps
- ...







Contact

You have a strong interest in **software quality assurance**? You are looking for a job as **working student** or working **full time**? You are looking for a topic for your **bachelor or master thesis**? Any other questions? Feel always free to contact us.

<https://jobs.cqse.eu/>



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94032 Passau

CQSE GmbH
Centa-Hafenbrädl-Straße 59
81249 München



Software Intelligence Night Passau

Software Quality in the Age Of Agentic Coding

 Thursday, July 9

 6 PM to 8:30 PM CEST

 CQSE, Steinweg 4, 94032 Passau

Register here: tmscl.me/passau267-t

